# Quiz 7- Solutions

MATH 3042- Quiz 7 Full Name:
Time: 20 minutes Set:

1. (7 marks) The following output is produced in R:

```
One Sample t-test

data:  memory_usage
t = 2.18, df = 24, p-value = 0.039

95 percent confidence interval:
 508.4  529.6
```

**Use for all questions as needed:**
qt(0.95,24) = 1.710882
qt(0.975,24) = 2.063899
qt(0.99,24) = 2.492159
qt(0.90,24) = 1.317836
qnorm(0.95) = 1.644854
qnorm(0.975) = 1.959964
qnorm(0.90) = 1.281552

Answer the following:

1. What sample mean was observed? $\bar{x} = \frac{508.4 + 529.6}{2} = 519$ MB

2. What sample size was used in this study? $24 + 1 = 25$

3. What is the sample standard deviation? $E = \frac{529.6 - 508.4}{2} = 10.6$

4. What is the 95% confidence interval for the true mean memory usage? $(508.4, 529.6)$ MB

Q3:

qt(0.975, 24) = 2.06389

$SE = \frac{10.6}{2.06389} = 5.136$

$5.136 = \frac{S}{\sqrt{25}}$

$\boxed{S = 25.68}$ MB

5. Write the statistically correct interpretation of this confidence interval.

If we repeatedly take samples of size 25 and construct the 95% CI for $\mu$ based on each sample set, 95% of those intervals will contain the true population mean.

6. Write the loosely correct interpretation of this confidence interval. we are 95% confident that true memory usage is between 508.4 to 529.6 MB.

7. Is 530 MB a plausible value for the true mean? Explain. This exceeds all values in the 95% CI. It is not a plausible value for the true mean.

2. (4 marks) A data center monitors packet corruption.

From 1500 packets, 72 were corrupted.

$n = 1500$ $\hat{p} = \frac{72}{1500} = 0.048$

$\hat{q} = 0.952$

a) Check the conditions for the Wald CI.
b) Construct the 95% Wald CI manually.
c) Engineers claim the corruption rate is below 6%. Does your CI support this claim?

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}\,\hat{q}}{n}}$$

a) $n\hat{p} = 1500 \times 0.048 = 72$ is greater than 10

$n\hat{q} = 1500 \times 0.952 = 1428$ is greater than 10

b) $Z_{\frac{\alpha}{2}} = \text{abs qnorm}(0.025) = 1.95996$ $SE = \sqrt{\frac{0.048 \times 0.952}{1500}} = 0.0055$

$E = (1.95996)(0.0055) = 0.0108$

$\hat{p} \pm E = 0.048 \pm 0.0108 = (0.0372, 0.0588)$

c)

3. A cybersecurity team wants to estimate the mean time (seconds) required to detect a malware signature. Previous measurements suggest the standard deviation is 4.2 seconds.

The team wants the estimate to have a margin of error no larger than 0.8 seconds with 95% confidence.

1. What minimum sample size is required?
2. If the team wants the margin of error reduced to 0.4 seconds, what sample size is required?

$$n = \left(\frac{Z_{\frac{\alpha}{2}} \cdot \sigma}{E}\right)^2$$

$\sigma = 4.2$

$\alpha = 5\% = 0.05$

$E = 0.8$

$Z_{\frac{\alpha}{2}} = 1.95996$

a) $n = \left(\frac{1.95996 \times 4.2}{0.8}\right)^2 = 106$

b) $n = \left(\frac{1.95996 \times 4.2}{0.4}\right)^2 = 424$